

Django + Gunicorn Setup on WSL

Windows Subsystem for Linux (WSL 2)

guide- WSL- Django project- **Gunicorn** run **NGINX** reverse proxy configure

1. WSL Environment Preparation

1.1 WSL Terminal Open

Windows- **WSL Terminal** open

WSL 2 recommended

WSL version check :

```
javascript.snippet JAVASCRIPT
wsl --status
```

Windows PowerShell :

```
javascript.snippet JAVASCRIPT
wsl -l -v
```

1.2 Python pip Check

WSL terminal-:

```
javascript.snippet JAVASCRIPT
python3 --version
```

```
javascript.snippet JAVASCRIPT
pip3 --version
```

Python pip installed :

```
javascript.snippet JAVASCRIPT
sudo apt updatesudo apt install python3 python3-pip python3-venv
```

2. Django Project Directory-

Windows- project location:

```
javascript.snippet JAVASCRIPT
C:\Users\Faruk\Desktop\myproject
```

WSL :

```
javascript.snippet JAVASCRIPT
cd /mnt/c/Users/Faruk/Desktop/myproject
```

Check :

```
javascript.snippet JAVASCRIPT
ls
```

files/folders :

```
javascript.snippet JAVASCRIPT
manage.pymyproject/
```

3. Virtual Environment

Project directory-:

```
javascript.snippet JAVASCRIPT
python3 -m venv venv
```

Virtual environment activate:

```
javascript.snippet JAVASCRIPT

source venv/bin/activate
```

Activate terminal- :

```
javascript.snippet JAVASCRIPT

(venv)
```

4. Required Packages Install

pip upgrade :

```
javascript.snippet JAVASCRIPT

pip install --upgrade pip
```

Django Unicorn:

```
javascript.snippet JAVASCRIPT

pip install django unicorn
```

project-requirements.txt ,:

```
javascript.snippet JAVASCRIPT

pip install -r requirements.txt
```

5. Django Configuration

settings.py- ALLOWED_HOSTS configure

Development/local testing- :

```
javascript.snippet JAVASCRIPT

ALLOWED_HOSTS = [ "127.0.0.1", "localhost", ]
```

WSL host/IP access , host- add

6. Database Migration

Project run migration :

```
javascript.snippet JAVASCRIPT
python manage.py migrate
```

7. Static Files Configuration

Production-style setup- settings.py:-

```
javascript.snippet JAVASCRIPT
STATIC_URL = "static/"
STATIC_ROOT = BASE_DIR / "staticfiles"
:
```

```
javascript.snippet JAVASCRIPT
python manage.py collectstatic
```

static files STATIC_ROOT directory- collect

:

```
javascript.snippet JAVASCRIPT
myproject/ manage.py myproject/ static/ staticfiles/ venv/
```

collectstatic production/deployment setup-

8. Gunicorn Test Run

Gunicorn Django application :

```
javascript.snippet JAVASCRIPT

gunicorn myproject.wsgi:application --bind 127.0.0.1:8000 --workers 3

:
```

```
javascript.snippet JAVASCRIPT

myproject
```

Django project- directory wsgi.py

Command- :

```
javascript.snippet JAVASCRIPT

gunicorn
```

Gunicorn server

```
javascript.snippet JAVASCRIPT

myproject.wsgi:application
```

Django WSGI application

```
javascript.snippet JAVASCRIPT

--bind 127.0.0.1:8000
```

Port 8000- server

```
javascript.snippet JAVASCRIPT

--workers 3
```

3 worker process

9. Browser Test

Windows browser- open :

```
javascript.snippet JAVASCRIPT
http://127.0.0.1:8000
```

```
javascript.snippet JAVASCRIPT
http://localhost:8000
```

Django application Gunicorn successfully

10. Gunicorn Background- Run

Terminal foreground- Gunicorn process

Simple background execution- :

```
javascript.snippet JAVASCRIPT
nohup gunicorn myproject.wsgi:application \ --bind 127.0.0.1:8000 \ --workers 3 \
> gunicorn.log 2>&1 &
```

```
javascript.snippet JAVASCRIPT
nohup
```

terminal session disconnect process

```
javascript.snippet JAVASCRIPT
&
```

command- background-

```
javascript.snippet JAVASCRIPT
> gunicorn.log
```

output gunicorn.log file-

```
javascript.snippet JAVASCRIPT
2>&1
```

error output- log file-

11. Gunicorn Process Check

Running Gunicorn process :

```
javascript.snippet JAVASCRIPT
ps aux | grep gunicorn
```

:

```
javascript.snippet JAVASCRIPT
pgrep -af gunicorn
```

12. Gunicorn Stop

PID :

```
javascript.snippet JAVASCRIPT
ps aux | grep gunicorn
```

:

```
javascript.snippet JAVASCRIPT
kill <PID>
```

:

```
javascript.snippet JAVASCRIPT
kill 1234
```

process :

```
kill -9 1234
```

13. Development Server Gunicorn

Django development- :

```
python manage.py runserver
```

Production-like testing- :

```
gunicorn myproject.wsgi:application --bind 127.0.0.1:8000 --workers 3
```

Difference

Server	Purpose
runserver	Django development
Gunicorn	WSGI application server
NGINX	Reverse proxy + static files

Development- NGINX

14. Optional: NGINX Setup

NGINX install :

```
sudo apt update
```



```
javascript.snippet JAVASCRIPT

sudo apt install nginx
```

Check:

```
javascript.snippet JAVASCRIPT

nginx -v
```

15. NGINX Configuration

Configuration file :

```
javascript.snippet JAVASCRIPT

sudo nano /etc/nginx/sites-available/myproject
```

Example configuration:

```
javascript.snippet JAVASCRIPT

server {    listen 80;    server_name localhost;    location /static/ {        alias /mnt/c/Users/Faruk/Desktop/myproject/staticfiles/;    }    location / {        proxy_pass http://127.0.0.1:8000;        proxy_set_header Host $host;        proxy_set_header X-Real-IP $remote_addr;        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;        proxy_set_header X-Forwarded-Proto $scheme;    }}

```

Important

Static files- alias path- STATIC_ROOT- match

:

```
javascript.snippet JAVASCRIPT

STATIC_ROOT = BASE_DIR / "staticfiles"
```

NGINX:

```
location /static/ {    alias /mnt/c/Users/Faruk/Desktop/myproject/staticfiles/;}
```

16. NGINX Site Enable

Symlink :

```
sudo ln -s /etc/nginx/sites-available/myproject \    /etc/nginx/sites-enabled/myproject
```

Default site conflict :

```
sudo rm /etc/nginx/sites-enabled/default
```

17. NGINX Configuration Test

Configuration check:

```
sudo nginx -t
```

Expected output- :

```
syntax is oktest is successful
```

18. NGINX Restart

WSL- systemd enabled :

```
javascript.snippet JAVASCRIPT

sudo systemctl restart nginx
```

Status:

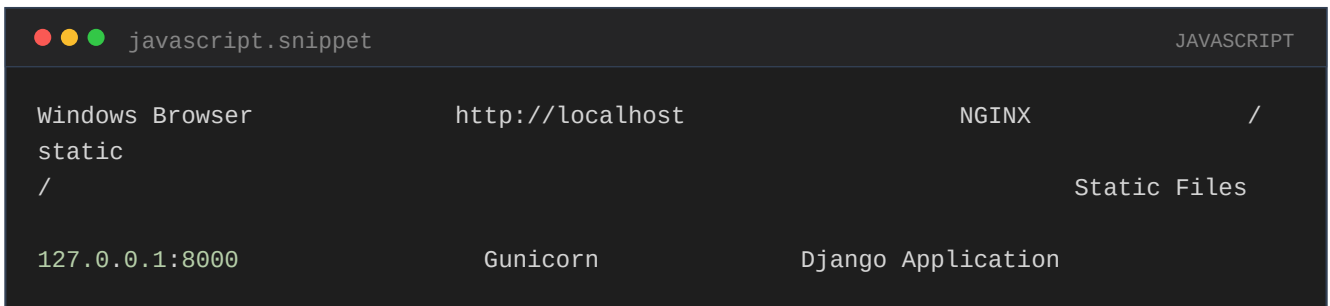
```
javascript.snippet JAVASCRIPT

sudo systemctl status nginx
```

systemd available , environment NGINX service start

19. Final Architecture

NGINX request flow :



20. Complete Basic Workflow

project :

Step 1 -- Project Directory

```
javascript.snippet JAVASCRIPT

cd /mnt/c/Users/Faruk/Desktop/myproject
```

Step 2 -- Virtual Environment

```
javascript.snippet JAVASCRIPT

source venv/bin/activate
```

Step 3 -- Run Gunicorn

```
javascript.snippet JAVASCRIPT
gunicorn myproject.wsgi:application \    --bind 127.0.0.1:8000 \    --workers 3
```

Step 4 -- Browser

Open:

```
javascript.snippet JAVASCRIPT
http://localhost:8000
```

21. Background Workflow

Background- :

```
javascript.snippet JAVASCRIPT
cd /mnt/c/Users/Faruk/Desktop/myproject
```

```
javascript.snippet JAVASCRIPT
source venv/bin/activate
```

```
javascript.snippet JAVASCRIPT
nohup gunicorn myproject.wsgi:application \    --bind 127.0.0.1:8000 \    --workers 3 \
> gunicorn.log 2>&1 &
```

Check:

```
javascript.snippet JAVASCRIPT
pgrep -af gunicorn
```

Log :

```
javascript.snippet JAVASCRIPT
tail -f gunicorn.log
```

Stop :

javascript.snippet

JAVASCRIPT

kill <PID>

22. Useful Commands Summary

Command	Purpose
source venv/bin/activate	Virtual environment activate
python manage.py migrate	Database migration
python manage.py collectstatic	Static files collect
python manage.py runserver	Django development server
gunicorn myproject.wsgi:application --bind 127.0.0.1:8000 --workers 3	Gunicorn run
ps aux grep gunicorn	Gunicorn process check
pgrep -af gunicorn	Gunicorn process check
kill <PID>	Gunicorn stop
tail -f gunicorn.log	Gunicorn log
sudo nginx -t	NGINX config test
sudo systemctl restart nginx	NGINX restart
sudo systemctl status nginx	NGINX status

23. Important Notes

1. WSL Development- useful

WSL Windows- Linux-based Django development environment

2. Gunicorn Windows-native server

Gunicorn Unix/Linux environment- WSL- Gunicorn practical

3. runserver Production Server

```
javascript.snippet JAVASCRIPT

python manage.py runserver
```

development-

4. Gunicorn + NGINX

deployment architecture:

```
javascript.snippet JAVASCRIPT

Client  NGINX  Gunicorn  Django  Database
```

5. WSL Production- consideration

Local development/test- WSL convenient Real production deployment- dedicated Linux server, VM cloud environment process management- systemd/supervisor

Quick Cheat Sheet

```
javascript.snippet JAVASCRIPT

# Go to projectcd /mnt/c/Users/Faruk/Desktop/myproject# Activate virtual environmentsource venv/bin/activate
```